Computational Intelligence: Overview

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Computational Intelligence (DS313/DS351) - 2026

About the Course ...

Course Information:

- Program on which the course is given: Bachelor of Computer & Information Sciences.
- Major element of program: Operations Research and Decision Support.
- Academic year/ level: 3rd year.
- Course code: DS313/DS351
- Lecturers: Assoc. Prof. Ayman Ghoneim and Dr. Mohamed Saad
- Lectures: 3 hrs / week (two 1.5 hrs lectures on Thursday)
- Practical: 1.5 hrs / week

About the lecturer ...

Hello !!!

Teaching and Learning Methods

Teaching and Learning Methods

1. Lecture slides.
2. Practical training and exercises.
3. Your "notes" in lectures/labs.

Material (reading material/slides/assignments/online lectures) will be uploaded to Google-Classroom:

Link:
https://classroom.google.com/c/ODQzNjMzMzUzNDMw?cjc=clqfydda
Code: clqfydda

I will use Google-Classroom to contact you.

Office hours: Tuesday 9 am to 12:30 pm (You need to confirm appointment by Email: a.ghoneim@fci-cu.edu.eg).

Assessment Methods

Student Assessment Methods and Weights

- Midterm Exam: 20% (give or take)
- Assignments / Quizzes / Project / Lab Test: 20 %
- Final Exam: 60%

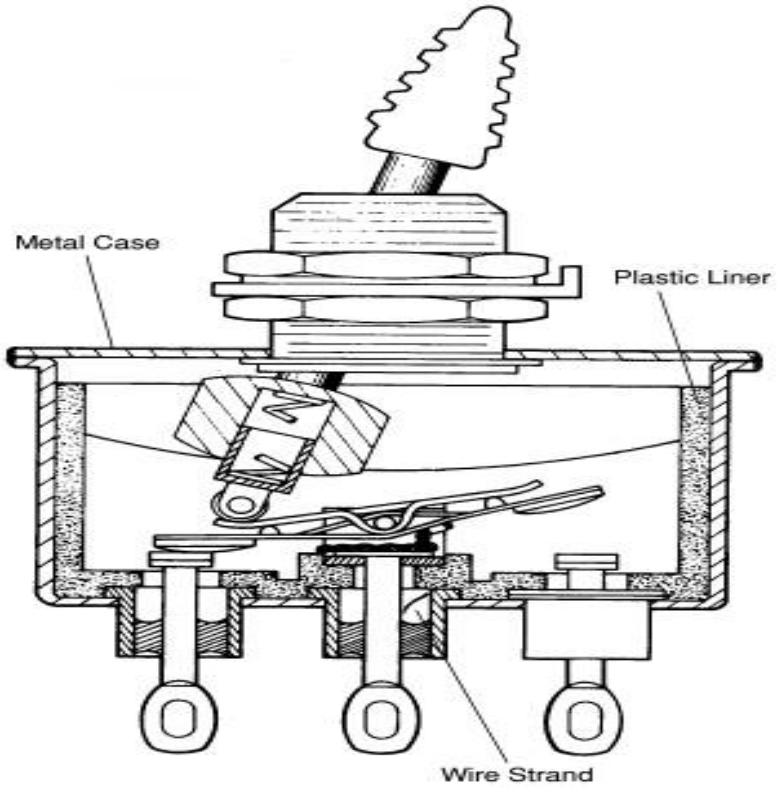
Code of Conduct

Code of Conduct:

- You are mature enough, step up for challenges (not only in this course).
- Attend your lectures/labs on time.
- Bring your laptops to the lab (and lectures if requested).
- Your Teaching Assistants to be treated with the utmost respect (not only for this course).

Flashback

First there was the Switch ...



Flashback

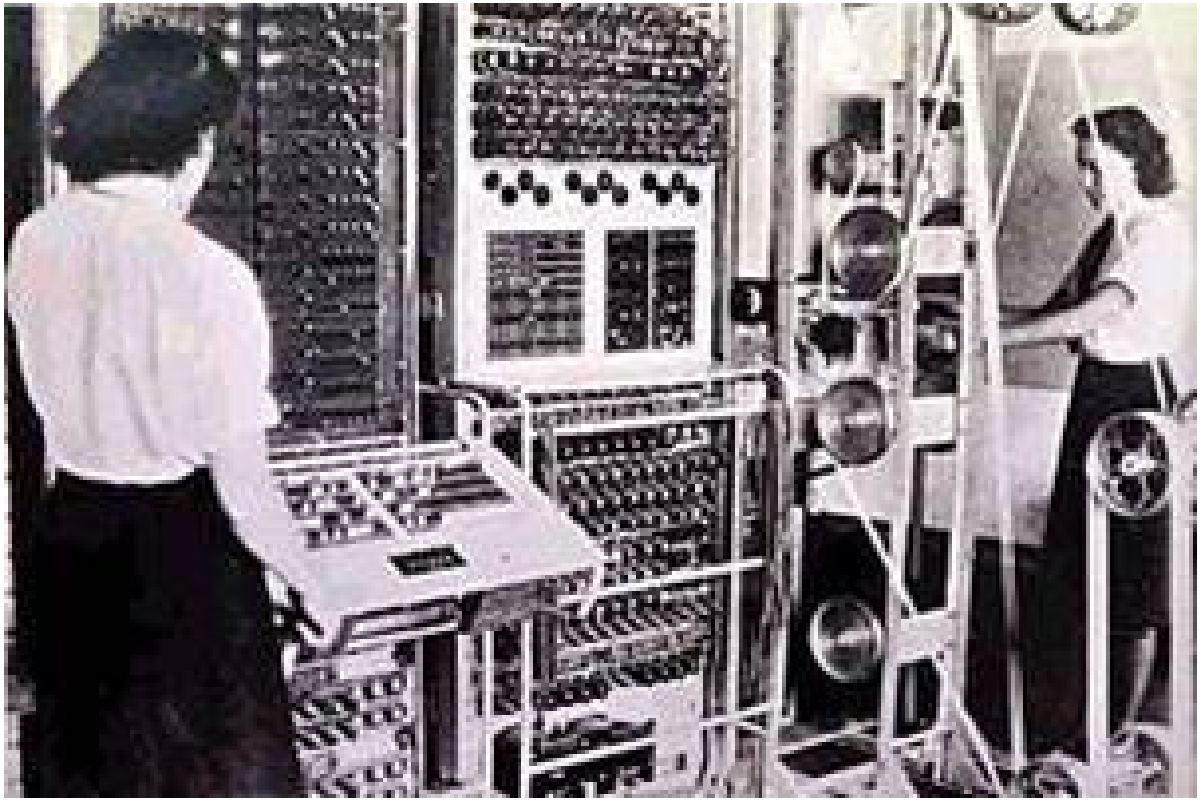
and something called a Computer ...

Flashback

Then there was the Thermionic Valve (an automated or controlled switch) ...

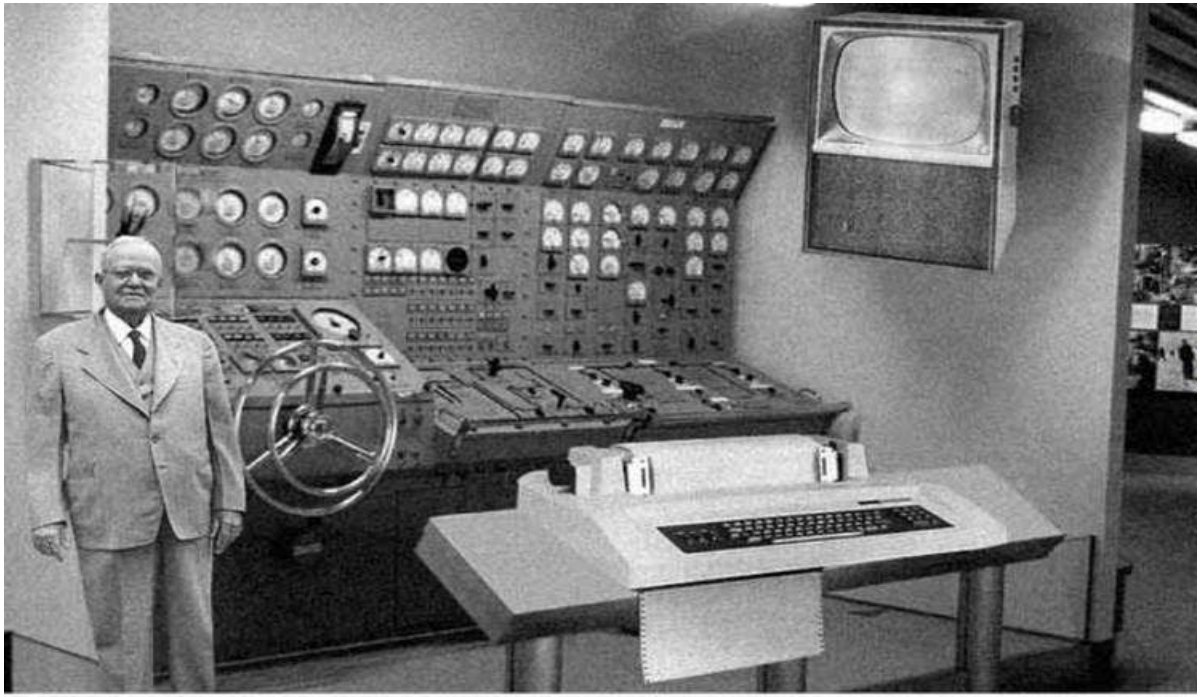
Flashback

and all sorts of things became possible !!!



Flashback

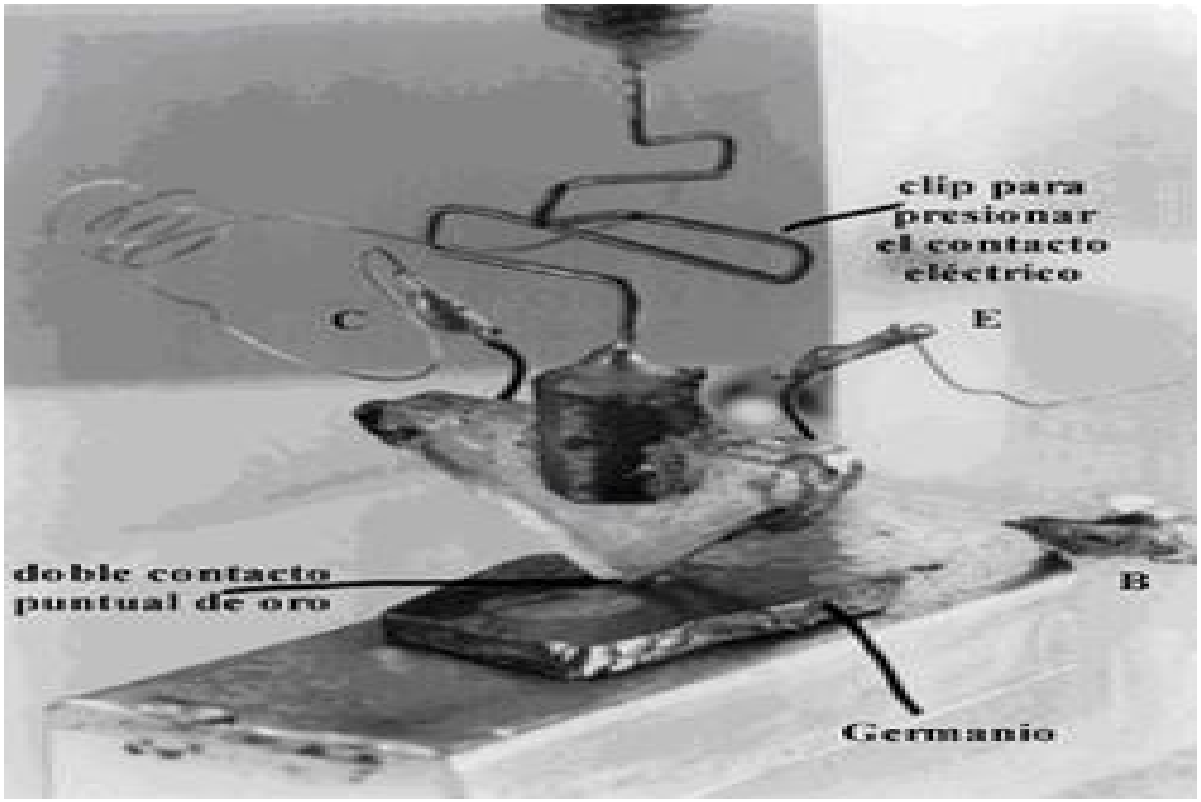
Some even dreamed of a "Home Computer"



Scientists from the RAND Corporation have created this model to illustrate how a "home computer" could look like in the year 2004. However the needed technology will not be economically feasible for the average home. Also the scientists readily admit that the computer will require not yet invented technology to actually work, but 50 years from now scientific progress is expected to solve these problems. With teletype interface and the Fortran language, the computer will be easy to use.

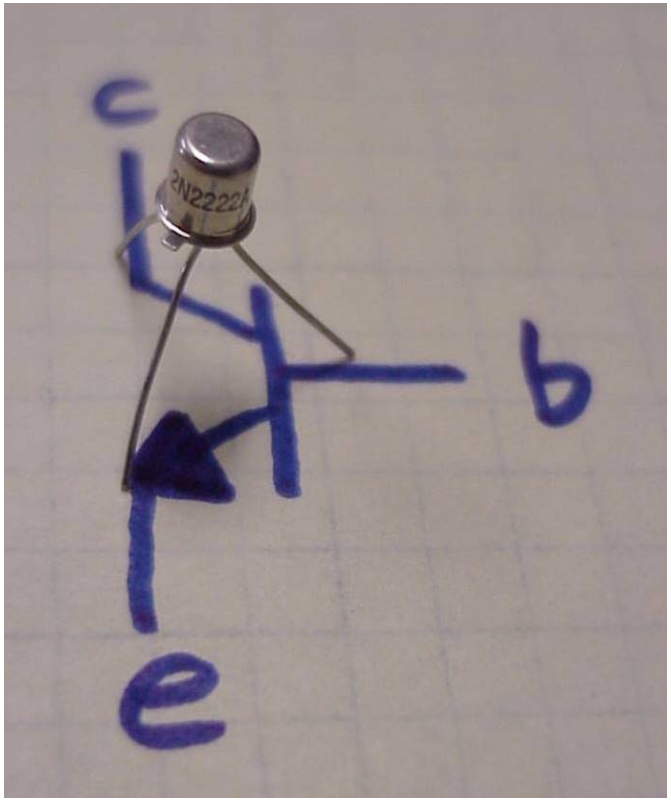
Flashback

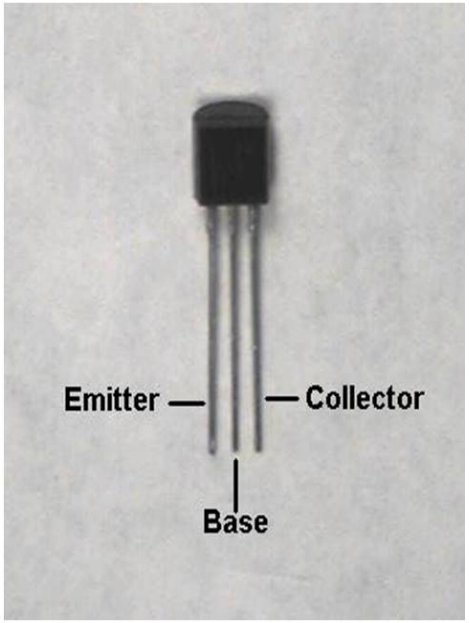
Then came the Transistor (another switch) ...



Flashback

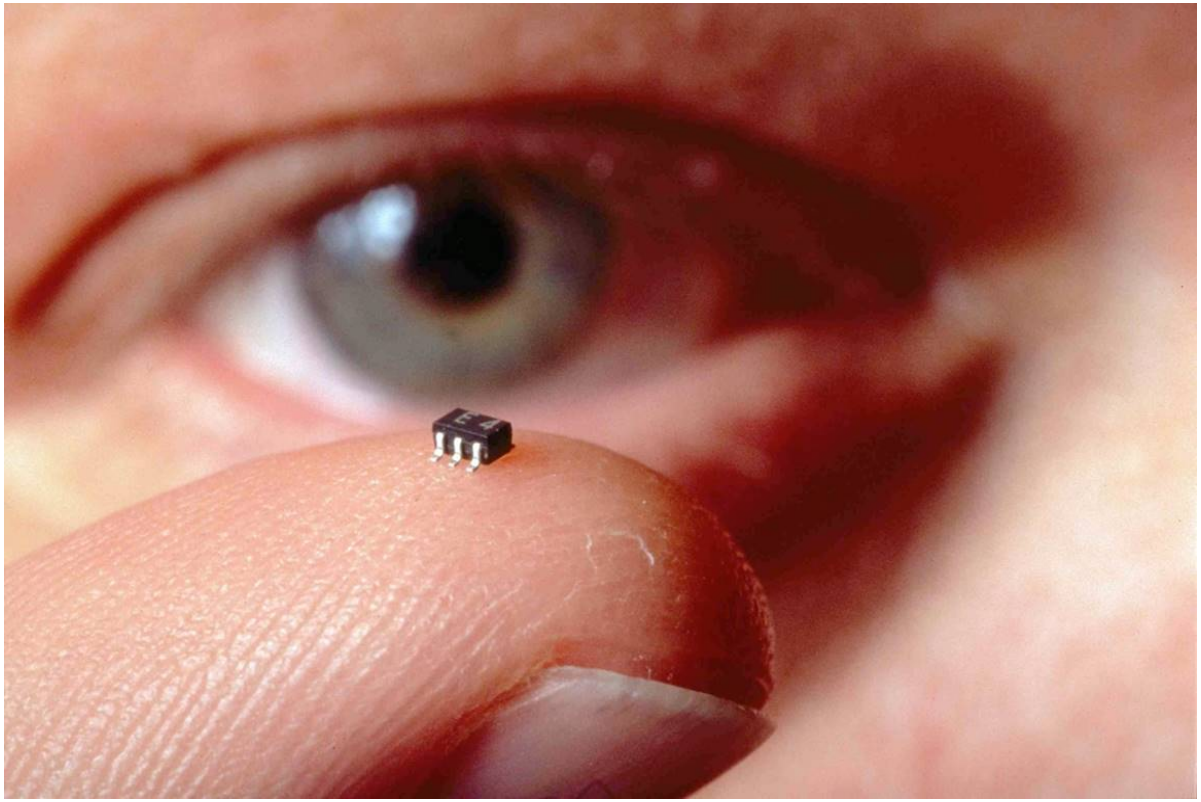
which got smaller ...





Flashback

and smaller, and smaller ...



Flashback

Very small logic gates (lots of them!) on a tiny chip enabled us to make adders, multipliers, decoders, storage devices.

Switches can repeat things many times and can control things, e.g., what sums to do then what to do next depending on the answer.

We can do lots of arithmetic really fast, so we called them "computers".

We can control switches, by telling them what to do in binary (a lot of hard work!!!)

So along came programming, we came up with mnemonics (assembly language) which were clearer to humans, and the switches decoded them into control signals.

Flashback

We used lots of patterns of these mnemonics over and over, and specifying every little step was still time-consuming and error-prone.

So Why not write "human-like" statements and get the switches to "compile" them into mnemonics (and hence into binary) for you? (Programming Languages)

Flashback

But ultimately they were still just binary switches.

Switches (programs) run into trouble/difficulty with:

- uncertainty, or missing information
- fuzzy concepts or categories
- noisy/erroneous information
- inexact, adaptive, abstract

Flashback

Yet humans and other animals do amazingly well in:

- Huge complexity
- Extraction of important features from masses of information
- Abstraction from detail
- dealing with uncertainty, inexactness, noise, incompleteness.

So how do we get computers to do these things?

Solution: "Intelligence", and it will be "artificial" by definition.

What is Intelligence?

Intelligence: (from psychology)

Judgment, otherwise called "good sense," "practical sense," "initiative," the faculty of adapting one's self to circumstances ... auto-critique, Alfred Binet (July 8, 1857 - October 18, 1911) was a French psychologist who invented the first practical intelligence test (An intelligence quotient (IQ); a total score derived from one of several standardized tests designed to assess human intelligence)

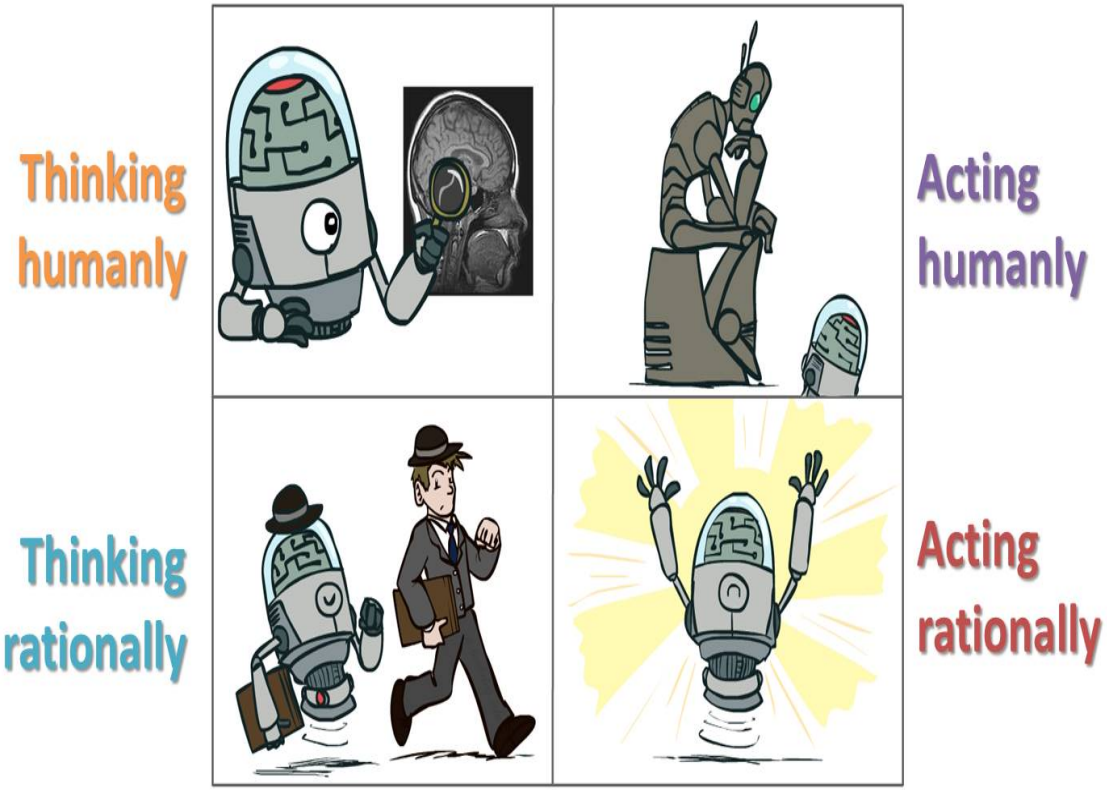
The resultant of the process of acquiring, storing in memory, retrieving, combining, comparing, and using in new contexts information and conceptual skills ... Lloyd G. Humphreys (December 12, 1913 - September 7, 2003) was an American psychologist

The capacity to learn and solve problems? (Websters dictionary) in particular,

- the ability to solve novel problems
- the ability to act rationally
- the ability to act like humans

What is Artificial Intelligence?

Artificial Intelligence (AI) can be classified into four categories.



What is Artificial Intelligence?

Systems that act like humans	Systems that think rationally
"The study of how to make computers do things at which, at the moment, people are better" (Rich and Knight, 1991) "The art of creating machines that perform functions that require intelligence when performed by people." (Kurzweil, 1990)	"The study of mental faculties through the use of computational models" (Charniack and McDermott, 1985). "The study of the computations that make it possible to perceive, reason, and act." (Winston, 1992)
Systems that think like humans	**Systems that act rationally**
"The automation of activities that we associate with human thinking, such as decision making, problem solving, learning" (Bellman, 1978) "The exciting new effort to make computers think ... machines with minds, in the full and literal sense." (Haugeland, 1985)	"AI .. is concerned with intelligent behavior in artifacts (Nilsson, 1998) "Computational Intelligence is the study of the design of intelligent agents." (Poole et al., 1998)

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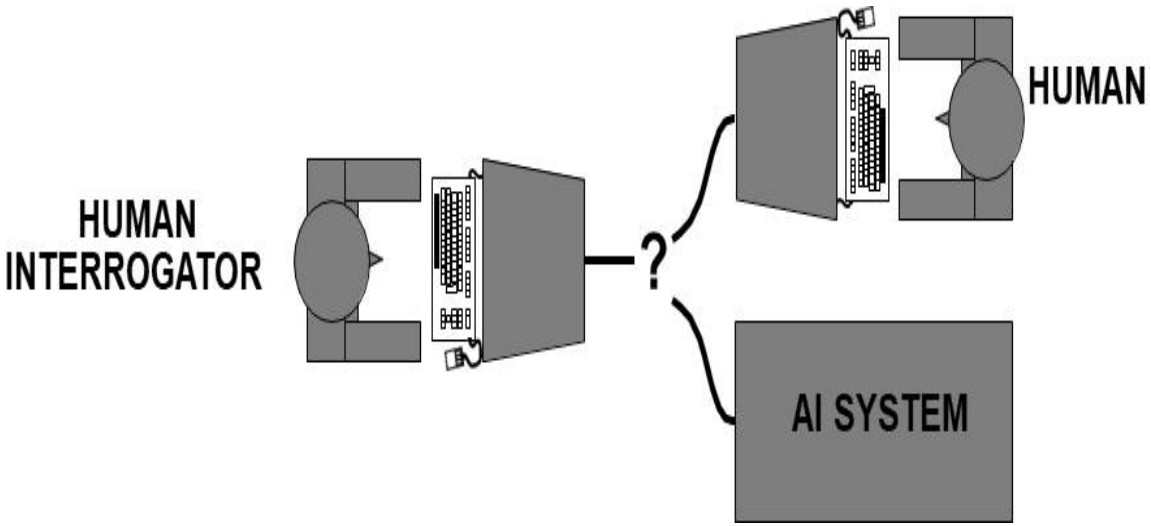
Systems that act like humans

The Turing test, the imitation game.

Turing (1950) "Computing machinery and intelligence":

- "Can machines think ?" or "Can machines behave intelligently ?"
- Operational test for intelligent behavior: the Imitation Game.



Systems that act like humans

Turing test (1950): Can a human interrogator tell whether (written) responses to her (written) questions come from a human or a machine? This requires:

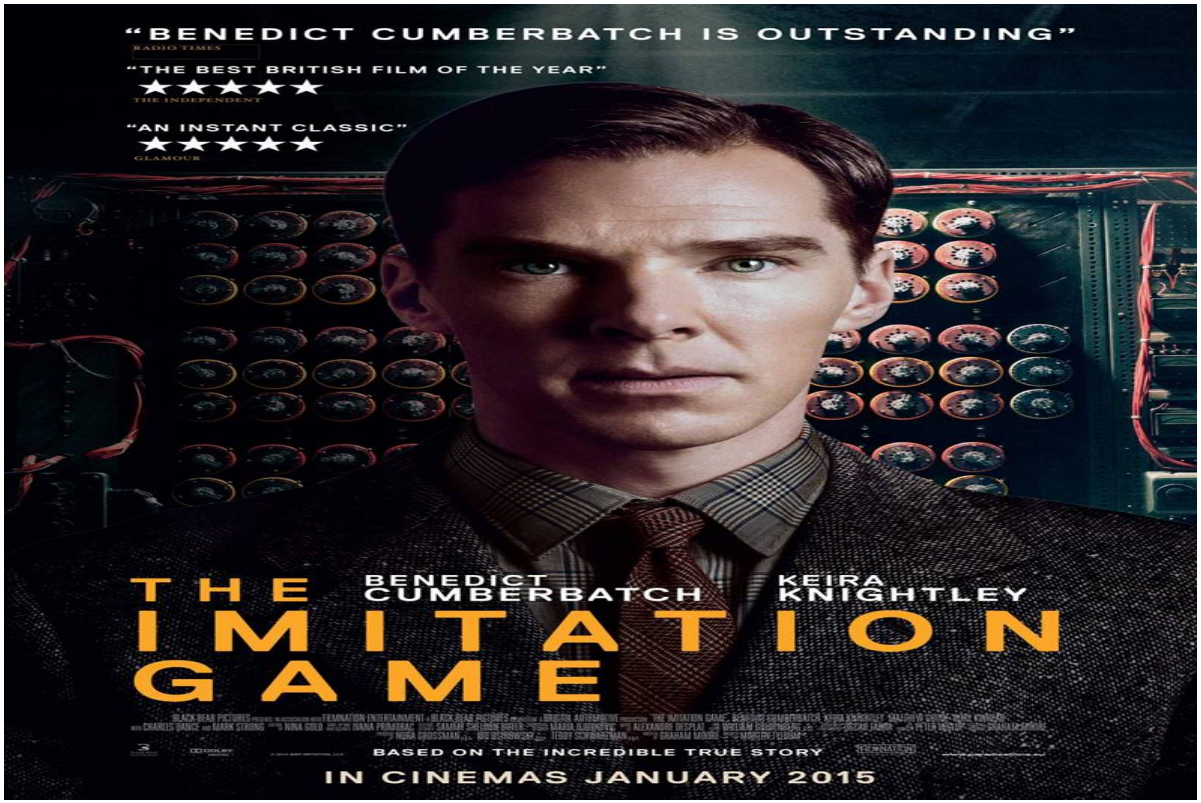
- Natural Language Processing
- Knowledge Representation
- Automated Reasoning
- Machine Learning

Total Turing Test (extended to include physical aspects of human behavior):

- Computer Vision
- Robotic

Why do we want an intelligent system to act like a human?
Because for many tasks, humans are still the Gold Standard.

Systems that act like humans



Chinese room argument

"Searle's thought experiment begins with this hypothetical premise: suppose that artificial intelligence research has succeeded in constructing a computer that behaves as if it understands Chinese. It takes Chinese characters as input and, by following the instructions of a computer program, produces other Chinese characters, which it presents as output. Suppose, says Searle, that this computer performs its task so convincingly that it comfortably passes the Turing test: it convinces a human Chinese speaker that the program is itself a live Chinese speaker. To all of the questions that the person asks, it makes appropriate responses, such that any Chinese speaker would be convinced that they are talking to another Chinese-speaking human being.?

The question Searle wants to answer is this: does the machine literally "understand" Chinese? Or is it merely simulating the ability to understand Chinese? Searle calls the first position "strong AI" and the latter "weak AI".

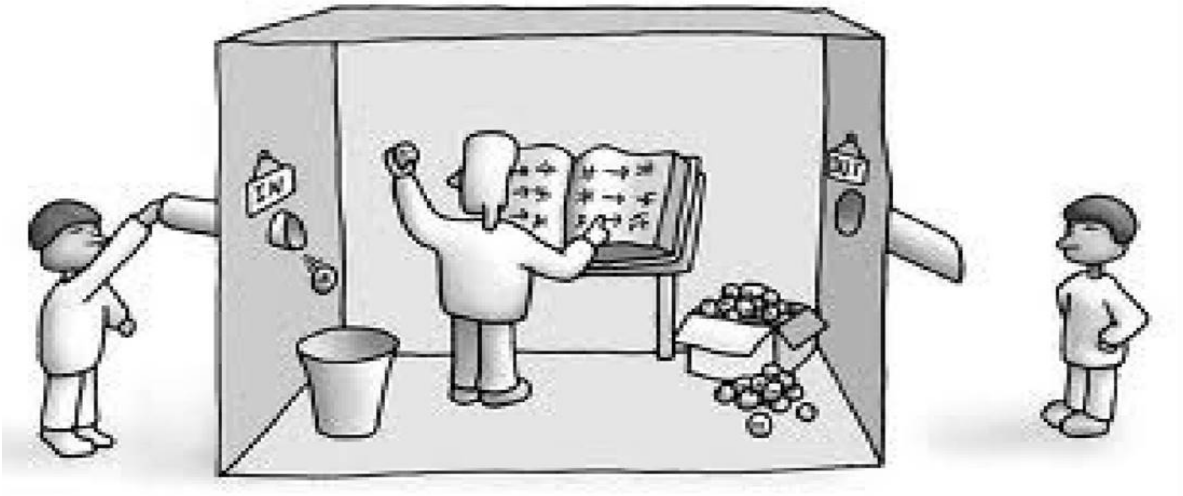
Chinese room argument

Searle then supposes that he is in a closed room and has a book with an English version of the computer program, along with sufficient paper, pencils, erasers, and filing cabinets. Searle could receive Chinese characters through a slot in the door, process them according to the program's instructions, and produce Chinese characters as output. If the computer had passed the Turing test this way, it follows, says Searle, that he would do so as well, simply by running the program manually. Searle asserts that there is no essential difference between the roles of the computer and himself in the experiment. Each simply follows a program, step-by-step, producing a behaviour which is then interpreted as demonstrating intelligent conversation. However, Searle would not be able to understand the conversation.

Searle argues that without "understanding" (or "intentionality"), we cannot describe what the machine is doing as "thinking" and since it does not think, it does not have a "mind" in anything like the normal sense of the word. Therefore, he concludes that "strong AI" is false.

Chinese room argument

If person inside does a great job of answering questions, can we say he or she understands? Even if she or he is only blindly following rules? (Of course the "person inside" is acting like an AI program)



WEAK AI Hypothesis: We can accurately simulate animal/human intelligence in a computer.

STRONG AI Hypothesis: We can create algorithms that are intelligent. (conscious ?)

Systems That Think Like Humans

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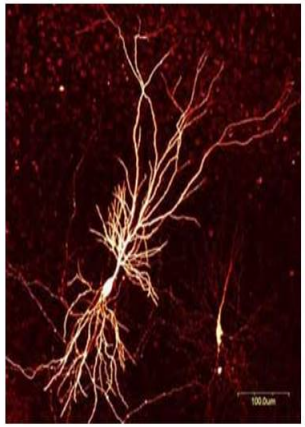
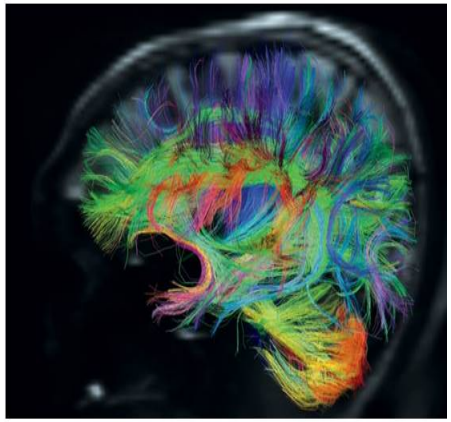
Systems That Think Like Humans

Need to study the brain as an information processing machine, in other words, use Computational Models to Understand the Actual Workings of Human Mind

- Devise/Choose a sufficiently precise theory of the mind.
- Express it as a computer program.
- Check match between program and human behavior (actions and timing) on similar tasks.

Tight connections with Cognitive Science and Neuroscience. Also known as descriptive approaches to AI.





Systems that think rationally

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Systems that think rationally

Logic: formalize idealized or right thinking, i.e. irrefutable reasoning processes. That is; patterns of argument that always yield correct conclusions when supplied with correct premises:

"Socrates is a man; all men are mortal; therefore Socrates is mortal."

Logistic tradition in AI aims to build computational frameworks based on logic, that is, describe problem in formal logical notation and apply general deduction procedures to solve it. Then use these frameworks to build intelligent systems.

Some examples are (Propositional Logic) and (Logic Programming). More advanced logic-based representations: Semantic Networks

Systems that think rationally

Main Research Problems/Challenges:

- Describing real-world problems and knowledge in logical notation.
- Proving Soundness and Completeness of various formalisms.
- How to represent often informal and uncertain domain knowledge and formalize it in logic notation (i.e., dealing with Uncertainty).
- Computational Complexity of finding a solution.
- A lot of "rational" behavior has nothing to do with logic.

Systems that act rationally

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Systems that act rationally

The "think rationally" approach focuses on correct inference.

But more is needed for rational behavior, e.g. How to behave when there is no provably correct thing to do (i.e. reasoning under uncertainty). Fully reactive behavior (instinct vs. reason).

Thinking vs Acting Rationally

AI with thinking rationally is know as "Traditional AI".

Traditional AI is strongly oriented to symbolic representations and manipulations (reasoning) in a top-down manner. That is, the structure of a given problem (environment, domain context) is analyzed beforehand and the construction of an intelligent system is based upon this structure.

One form of AI with acting rationally is know as "Computational Intelligence" (CI).

Thinking vs Acting Rationally

Traditional AI - Key Features

- Top-down
- Knowledge based (e.g. Expert systems).
- Mainly dependent on symbolic manipulation, e.g.,
  - If all fruits are sugary $\Rightarrow$ given,
  - If apple is a fruit $\Rightarrow$ user input,
  - Then apple is sugary $\Rightarrow$ system conclusion.
- Rigid

Thinking vs Acting Rationally

CI - Key Features

- Bottom-up
- Adaptive and flexible
- Works on numeric data to infer symbols.
- CI techniques (i.e., evolutionary, fuzzy, and neurocomputing) model natural processes or end-products associated with intelligent behaviour, either at the level of neuronal activity and function, human behaviour, or evolutionary learning in the terms of adaptive behaviour or adaptive genetics.
- Only by pushing it to the extreme, from these premises it may be implied that (traditional) AI systems are not intelligent, while CI systems are.

Computational Intelligence - Associated terms

The following are associated terms with Computational Intelligence.

Natural-Computation:

The use of nature inspired approaches to problem solving (e.g. evolutionary, neural, DNA, and quantum computing).

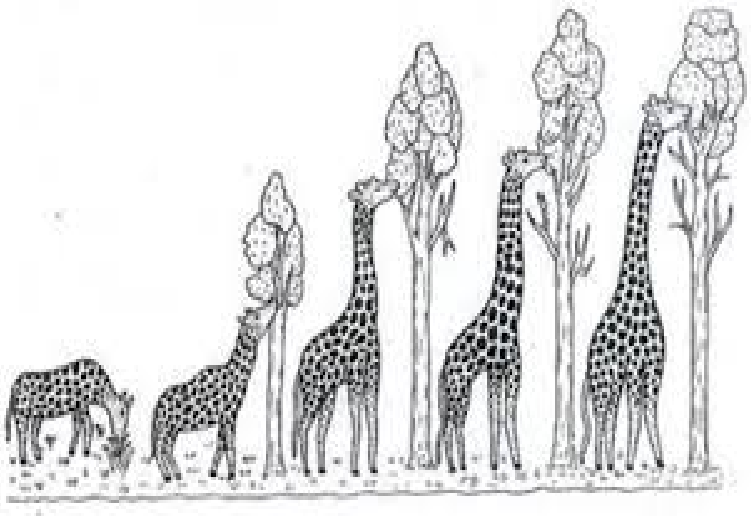
Soft Computing

OR algorithms usually come with crisp (and strict) conditions on the scope of applicability and proven guarantees for a solution (or even an optimal solution), soft computing puts no conditions on the problem but also provides no guarantees for success, a deficiency which is compensated by the robustness of the methods.

Computational Intelligence - Paradigms

Evolutionary Computations:

Uses Darwinistic methods for generating and evaluating a population of possible solutions to a problem. It is known for its generality and robustness.



Computational Intelligence - Paradigms

Swarm Intelligence:

Models of collective behavior of natural swarms (ants, bees, termites, etc) are seen as templates for problem solving.

Computational Intelligence - Paradigms

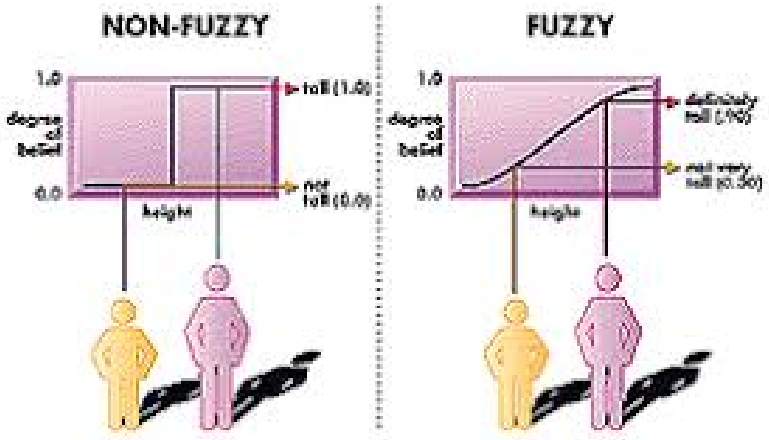
Neural Computations:

uses the (human) brain as a model for solving problems. It is known for its ability to deal with noisy and variable information. (e.g. stock market prediction, forecasting, and classification)

Computational Intelligence - Paradigms

Fuzzy Computations:

Instead of using Boolean logic, Fuzzy Systems use the notions of almost true or false to solve problems were Boolean logic would fail.



Why Computational Intelligence?

Why CI?

- Optimization
- Learning and classification
- Building autonomous systems
- Simulating collective behavior
- Understanding complex adaptive systems

Course Objectives

This course will enable you to:

- Understand and develop evolutionary algorithms for optimizing solutions to real-world problems, including the optimization of neural and fuzzy systems.
- Understand and develop fuzzy membership functions and combine them for use in relating fuzzy sets.